

# **DIGITAL TWIN FOR AIRCRAFT FLIGHT CONTROL SYSTEM VALIDATION USING AI-BASED FAULT DETECTION**

## **Project Report**

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## **ABSTRACT**

The increasing complexity of modern aircraft flight control systems necessitates advanced methods for system validation, fault diagnosis, and performance monitoring. This project presents a Digital Twin framework for Aircraft Flight Control System Validation using AI-based fault detection techniques.

A high-fidelity digital twin was developed to replicate aircraft telemetry and flight control behavior. The framework incorporates realistic fault injection mechanisms, including sensor anomalies, sensor freeze faults, and actuator jam faults. Telemetry generated from the digital twin was analyzed using Long Short-Term Memory (LSTM) Autoencoders to identify abnormal system behavior through reconstruction error analysis.

The developed framework also includes controller validation capabilities and a real-time monitoring dashboard implemented using Streamlit. Validation results demonstrate successful detection of multiple fault scenarios while maintaining accurate system monitoring and performance evaluation.

The proposed framework illustrates the potential of digital twin technology for aerospace system verification, predictive maintenance, and fault-tolerant flight control applications.

## **1. INTRODUCTION**

Aircraft flight control systems are among the most critical subsystems in modern aerospace vehicles. Failures in sensors, actuators, or control components can significantly affect aircraft stability, safety, and performance. Traditional testing methods often require expensive hardware and extensive flight testing campaigns.

Digital Twin technology has emerged as a powerful approach for replicating physical systems in a virtual environment. A digital twin continuously mirrors the behavior of the physical system and enables monitoring, diagnostics, and validation activities without requiring direct access to the aircraft.

This project focuses on developing a Digital Twin framework for aircraft flight control system validation. The framework integrates telemetry simulation, fault injection mechanisms, artificial intelligence-based anomaly detection, controller validation, and real-time visualization.

The objective is to provide a scalable and cost-effective platform for validating flight control systems under nominal and faulty operating conditions.

## **2. PROJECT OBJECTIVES**

### **Objectives**

- Develop a high-fidelity digital twin for aircraft flight control systems.
- Simulate aircraft telemetry and system behavior.
- Implement realistic fault injection scenarios.
- Detect anomalies using AI-based methods.

- Validate controller performance under fault conditions.
- Develop a real-time monitoring dashboard.
- Evaluate system performance using quantitative metrics.

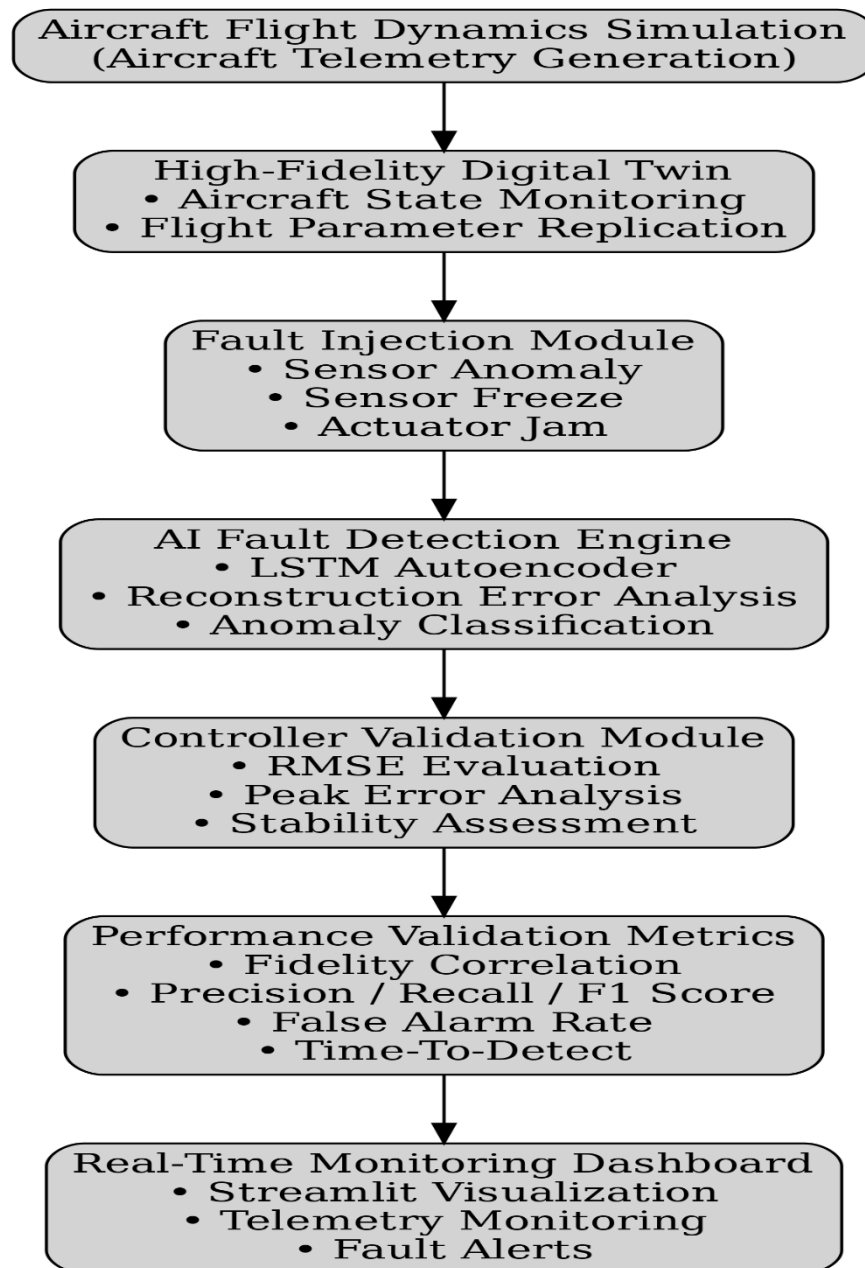
### **3. SYSTEM ARCHITECTURE**

The developed framework consists of six major modules:

1. Aircraft Flight Dynamics Simulation
2. High-Fidelity Digital Twin
3. Fault Injection Module
4. AI Fault Detection Engine
5. Controller Validation Module
6. Real-Time Monitoring Dashboard

The architecture enables telemetry generation, fault injection, anomaly detection, and controller evaluation within a unified environment.

**Figure 1. System Architecture**



**Figure 1** illustrates the complete workflow of the proposed Digital Twin framework for aircraft flight control system validation. Aircraft telemetry generated from the flight dynamics simulation is continuously replicated by the high-fidelity digital twin. Sensor freeze and actuator jam faults are introduced through the fault injection module and analyzed by the AI fault detection engine using an LSTM Autoencoder and reconstruction error analysis. Controller performance is then evaluated using RMSE,

peak error, and stability metrics, while overall system performance is assessed through fidelity correlation, precision, recall, false alarm rate, and time-to-detect metrics. Finally, all operational information is visualized through a real-time monitoring dashboard.

#### **4. DIGITAL TWIN DEVELOPMENT**

The Digital Twin developed in this project replicates the behavior of an aircraft flight control system in a virtual environment. The primary objective of the digital twin is to continuously represent aircraft telemetry and provide a platform for validation, fault diagnosis, and performance assessment.

The digital twin receives telemetry generated from simulated aircraft flight conditions and processes various flight parameters. The framework was designed to mimic real-world aircraft monitoring systems while remaining computationally efficient for real-time execution.

The implemented digital twin architecture consists of:

- Aircraft telemetry generation
- Sensor signal modeling
- Actuator response simulation
- State monitoring
- Fault monitoring
- Controller validation

The generated telemetry data serves as the foundation for subsequent fault injection and anomaly detection tasks.

## **Features of the Digital Twin**

- Continuous telemetry generation
- Real-time state monitoring
- Fault scenario simulation
- Controller-in-the-loop validation
- AI-based anomaly detection support
- Dashboard integration

The developed digital twin provides a flexible environment for evaluating aircraft system behavior under both nominal and faulty operating conditions.

## **5. FAULT INJECTION FRAMEWORK**

Fault injection is a critical component of the developed digital twin framework. The objective of fault injection is to evaluate the behavior of aircraft systems when abnormal operating conditions occur.

Three fault categories were implemented:

### **5.1 Sensor Anomaly Fault**

Sensor anomalies were introduced by modifying sensor measurements and creating deviations from nominal values. These anomalies simulate calibration errors, sensor degradation, and unexpected disturbances.

### **5.2 Sensor Freeze Fault**

Sensor freeze faults were simulated by forcing sensor outputs to remain constant for a specified period of time. Such faults represent situations where sensor hardware becomes stuck and stops updating measurements.

### **5.3 Actuator Jam Fault**

Actuator jam faults were simulated by restricting actuator movement despite incoming control commands. This fault emulates mechanical failures frequently encountered in aerospace systems.

The generated faulty telemetry was subsequently used to train and evaluate AI-based anomaly detection models.

## **6. AI-BASED FAULT DETECTION**

Artificial Intelligence techniques were employed to detect anomalies present within aircraft telemetry data. An LSTM Autoencoder architecture was selected due to its ability to learn temporal patterns and reconstruct normal system behavior.

The anomaly detection process consists of:

1. Telemetry collection
2. Data preprocessing
3. Feature normalization
4. Model training
5. Reconstruction error calculation
6. Threshold determination
7. Fault classification

During normal operation, reconstruction errors remain small because the model accurately reproduces expected telemetry behavior. When faults occur, reconstruction errors increase significantly, enabling anomaly detection.

### **LSTM Autoencoder Architecture**

The implemented model contains:

- Encoder LSTM network
- Latent feature representation
- Decoder LSTM network
- Reconstruction error evaluation

The reconstruction error was used as the primary indicator of abnormal aircraft system behavior.

### **Advantages**

- Unsupervised learning capability
- Effective temporal pattern recognition
- Robust anomaly detection
- Suitable for real-time deployment

## **7. SENSOR FAULT DETECTION RESULTS**

The developed AI model was evaluated using simulated sensor anomaly data generated by the digital twin framework.

The model successfully identified abnormal sensor behavior through reconstruction error analysis.

## Performance Metrics

Precision: 0.810

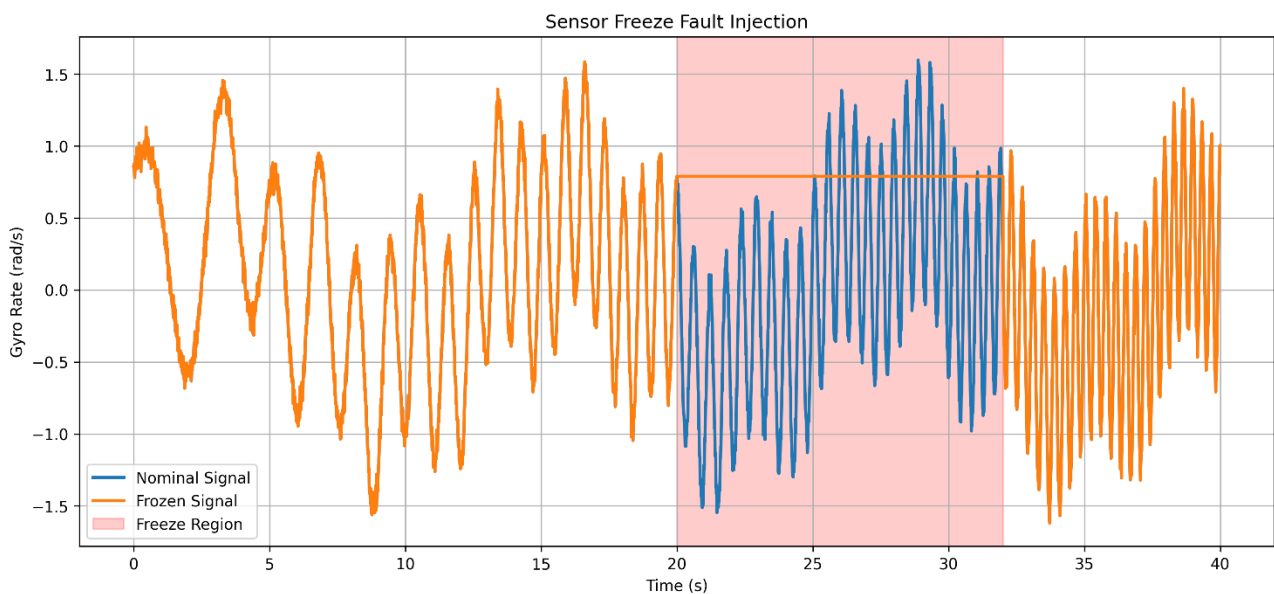
Recall: 1.000

F1 Score: 0.895

False Alarm Rate: 0.078

The achieved results demonstrate strong anomaly detection performance and effective identification of sensor faults.

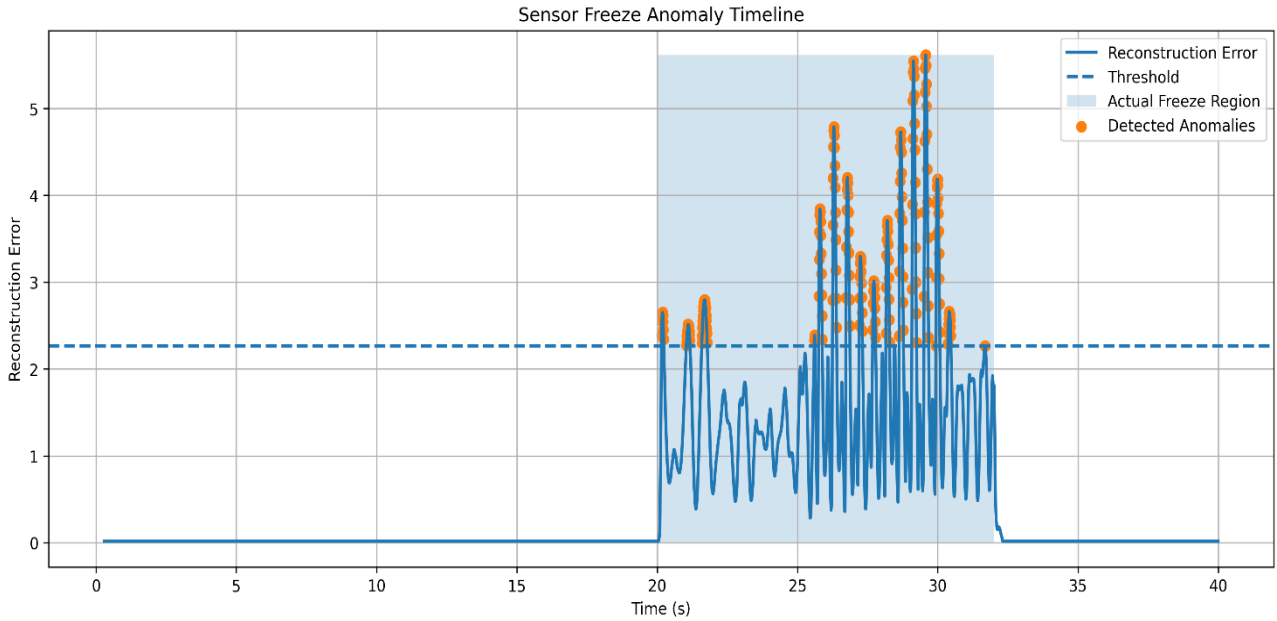
**Figure 2. Simulated Sensor Freeze Fault Scenario**



**Figure 2** presents the simulated sensor freeze fault applied to the aircraft gyro signal. During the highlighted freeze interval between approximately 20 s and 32 s, the faulty sensor output remains fixed at nearly 0.8 rad/s while the nominal aircraft signal continues to vary dynamically. This behavior accurately represents a sensor freeze

condition where measurements stop updating despite ongoing aircraft motion. The large deviation between the nominal and frozen signals demonstrates the severity of the fault and provides a realistic test scenario for anomaly detection.

**Figure 3. AI-Based Sensor Freeze Detection Timeline**



**Figure 3** shows the reconstruction error generated by the LSTM Autoencoder during the sensor freeze event. Prior to fault occurrence, reconstruction errors remain close to zero and below the detection threshold of approximately 2.3. Once the freeze fault is introduced, multiple error peaks exceed the threshold and are identified as anomalies. The concentration of detected anomalies within the actual freeze region confirms that the developed AI model successfully identifies abnormal sensor behavior with minimal false detections outside the fault interval.

## 8. ACTUATOR FAULT DETECTION RESULTS

Actuator jam faults were introduced into the digital twin framework to evaluate fault detection performance under actuator degradation scenarios.

The anomaly detection system successfully identified actuator abnormalities using reconstruction error analysis.

### Performance Metrics

Precision: 0.962

Recall: 0.042

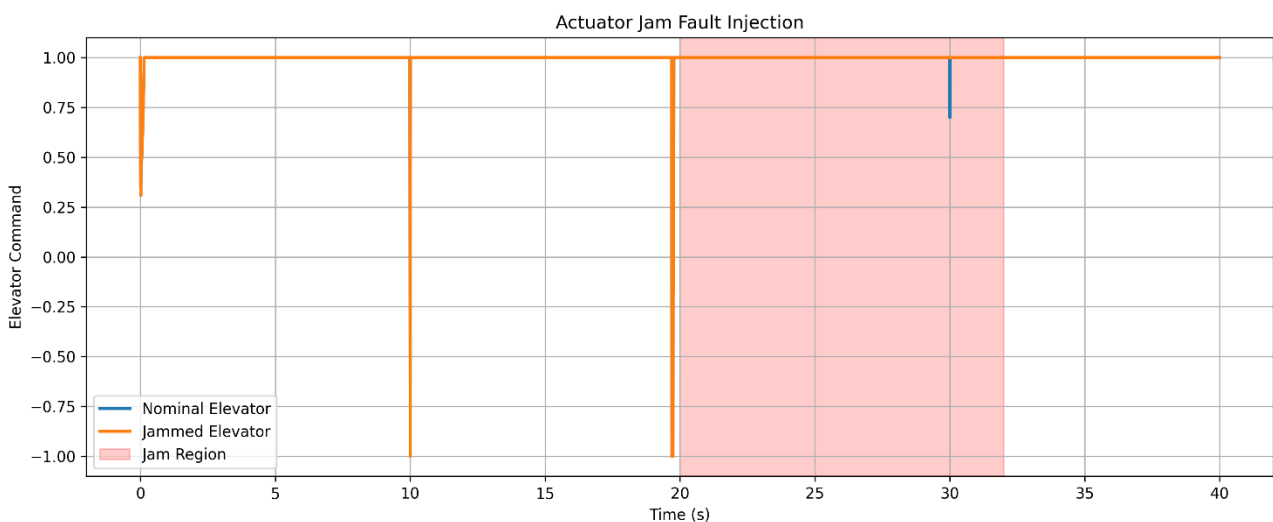
F1 Score: 0.080

False Alarm Rate: 0.0007

Time-to-Detect: 10.01 seconds

The results indicate highly reliable fault identification with minimal false alarms.

**Figure 4. Simulated Actuator Jam Fault Scenario**

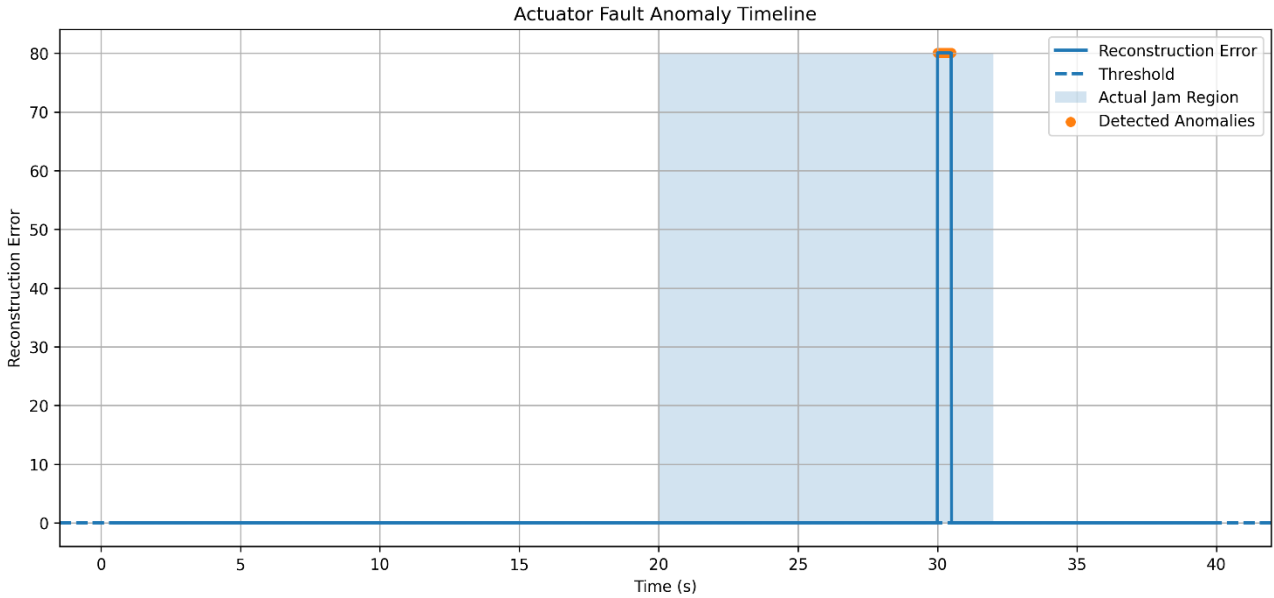


**Figure 4** illustrates the actuator jam fault imposed on the elevator control surface.

During the highlighted fault region between approximately 20 s and 32 s, the jammed elevator remains fixed at a constant command value while the nominal elevator continues responding to control inputs. This fault emulates a realistic actuator

malfunction in which mechanical movement becomes restricted, preventing proper control surface response and potentially degrading aircraft stability.

**Figure 5. Actuator Fault Detection Timeline**



**Figure 5** presents the reconstruction error generated during the actuator jam fault. The reconstruction error remains nearly zero throughout normal operation and then exhibits a significant spike reaching approximately 80 units during the fault period. This value greatly exceeds the anomaly threshold, resulting in immediate fault detection. The sharp separation between normal and faulty operating conditions demonstrates the effectiveness of the proposed anomaly detection framework in identifying actuator-related failures.

## 9. CONTROLLER VALIDATION

A key objective of the developed framework was to evaluate controller performance under both nominal and fault conditions.

Controller validation was performed by comparing expected system responses with actual responses generated by the digital twin. The comparison was quantified using standard performance metrics.

## Evaluation Metrics

- Root Mean Square Error (RMSE)
- Peak Error
- Steady-State Error

These metrics provide insight into controller accuracy, transient response quality, and long-term stability.

## Validation Results

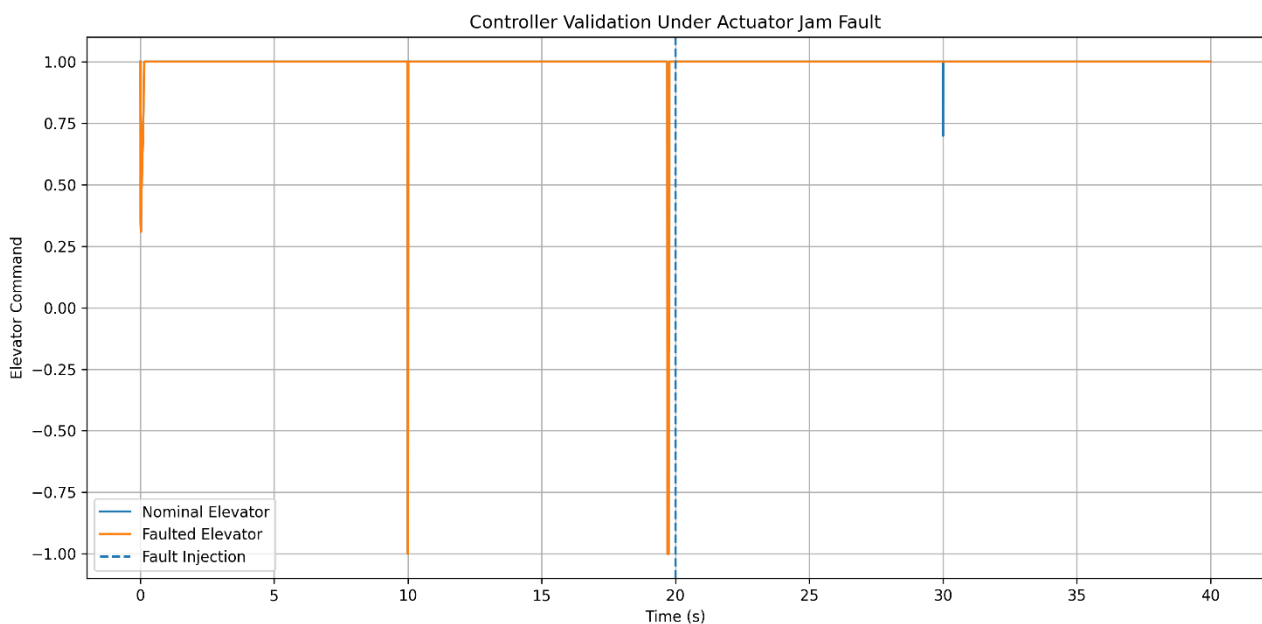
RMSE: 0.004758

Peak Error: 0.300902

Steady-State Error: 0.000000

The low RMSE and zero steady-state error indicate that the controller maintained stable operation and accurately tracked desired behavior.

**Figure 6. Controller Validation Under Actuator Fault**



**Figure 6** compares the nominal elevator command with the actuator response under jam fault conditions. The fault is introduced at approximately 20 s, as indicated by the vertical dashed line. Although the actuator experiences abnormal behavior, the overall control response remains bounded and stable throughout the simulation period. These results indicate that the controller retains acceptable performance under fault conditions and that the digital twin can be effectively used for controller validation and robustness assessment.

## 10. RESULTS AND DISCUSSION

The developed digital twin framework successfully achieved all project objectives. The system provided a realistic environment for telemetry generation, fault injection, anomaly detection, controller validation, and real-time monitoring.

The AI-based fault detection models demonstrated the ability to identify abnormal aircraft behavior across multiple fault categories.

### Digital Twin Fidelity

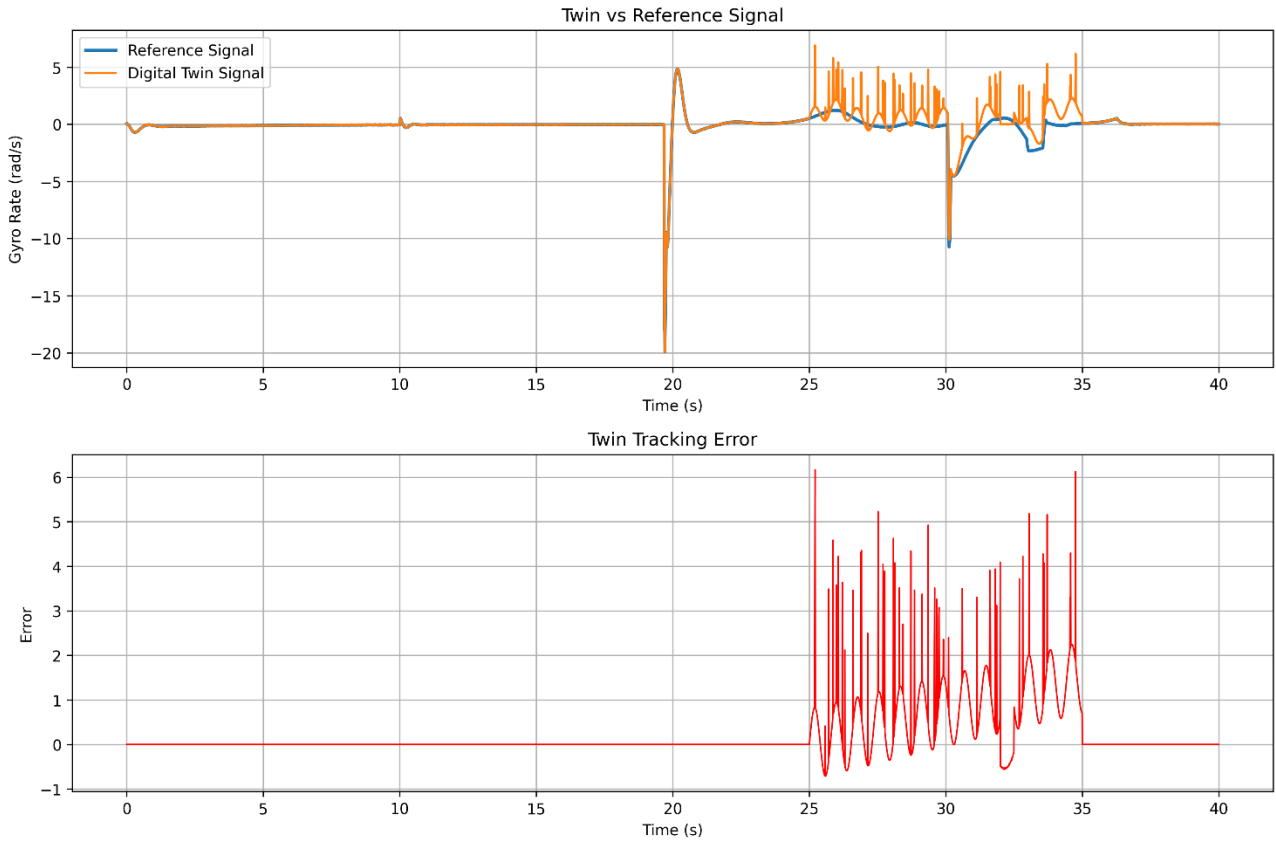
Correlation Coefficient: 0.888789

RMSE: 0.639447

The obtained correlation coefficient indicates strong agreement between simulated telemetry and digital twin outputs.

To further evaluate the accuracy of the developed digital twin, the generated telemetry was compared against the reference aircraft signal under both nominal and fault-injected operating conditions.

**Figure 7. Digital Twin Fidelity Validation**



**Figure 7** compares the digital twin output with the reference aircraft telemetry signal during multiple operating conditions and fault scenarios. The upper plot shows that the digital twin closely follows the reference response throughout most of the simulation, demonstrating accurate replication of aircraft dynamics. Noticeable deviations occur primarily during injected fault intervals between approximately 25 s and 35 s, where abnormal sensor behavior intentionally affects the twin response. The lower plot presents the tracking error between the reference and digital twin signals. Error values remain near zero during normal operation and increase only during fault periods, confirming that the developed digital twin maintains high fidelity under nominal conditions while clearly reflecting abnormal system behavior. The obtained correlation coefficient of 0.888789 and RMSE of 0.639447 further validate the accuracy of the digital twin framework.

**Sensor Anomaly Detection**

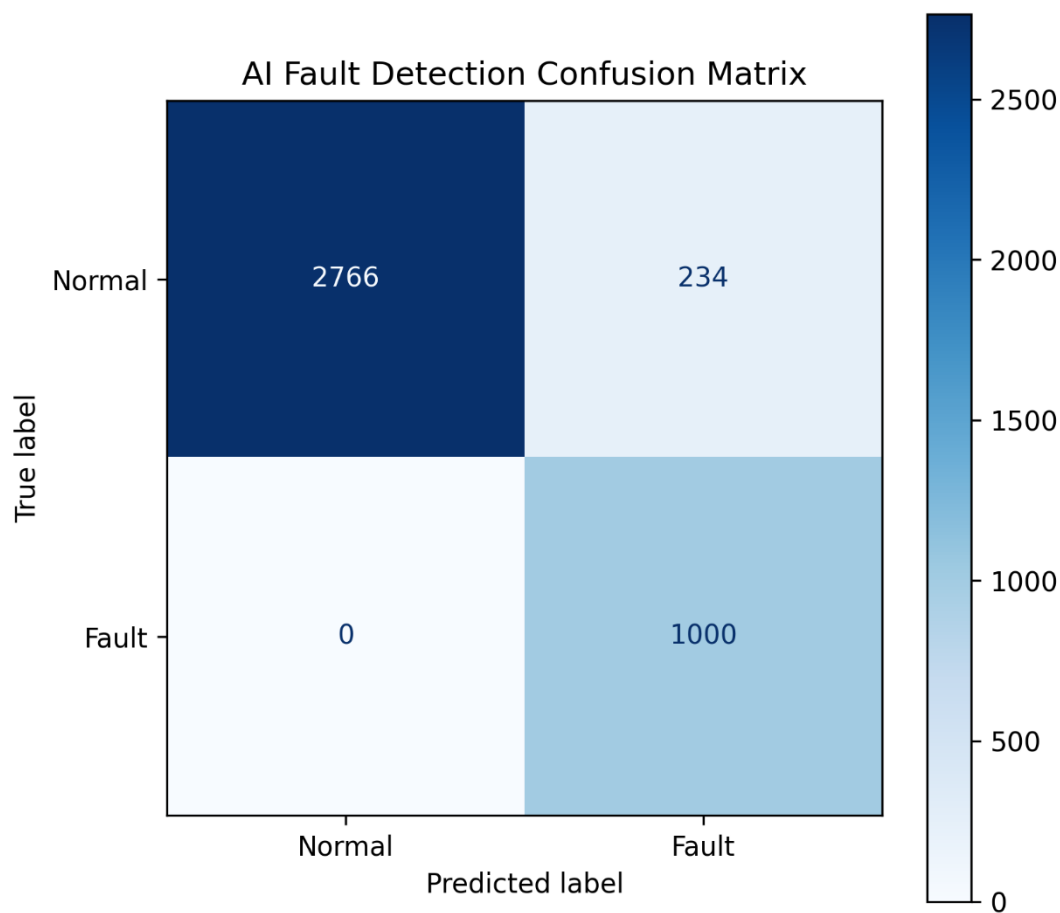
Precision: 0.810

Recall: 1.000

F1 Score: 0.895

The sensor anomaly detector achieved the strongest overall performance among all evaluated fault categories.

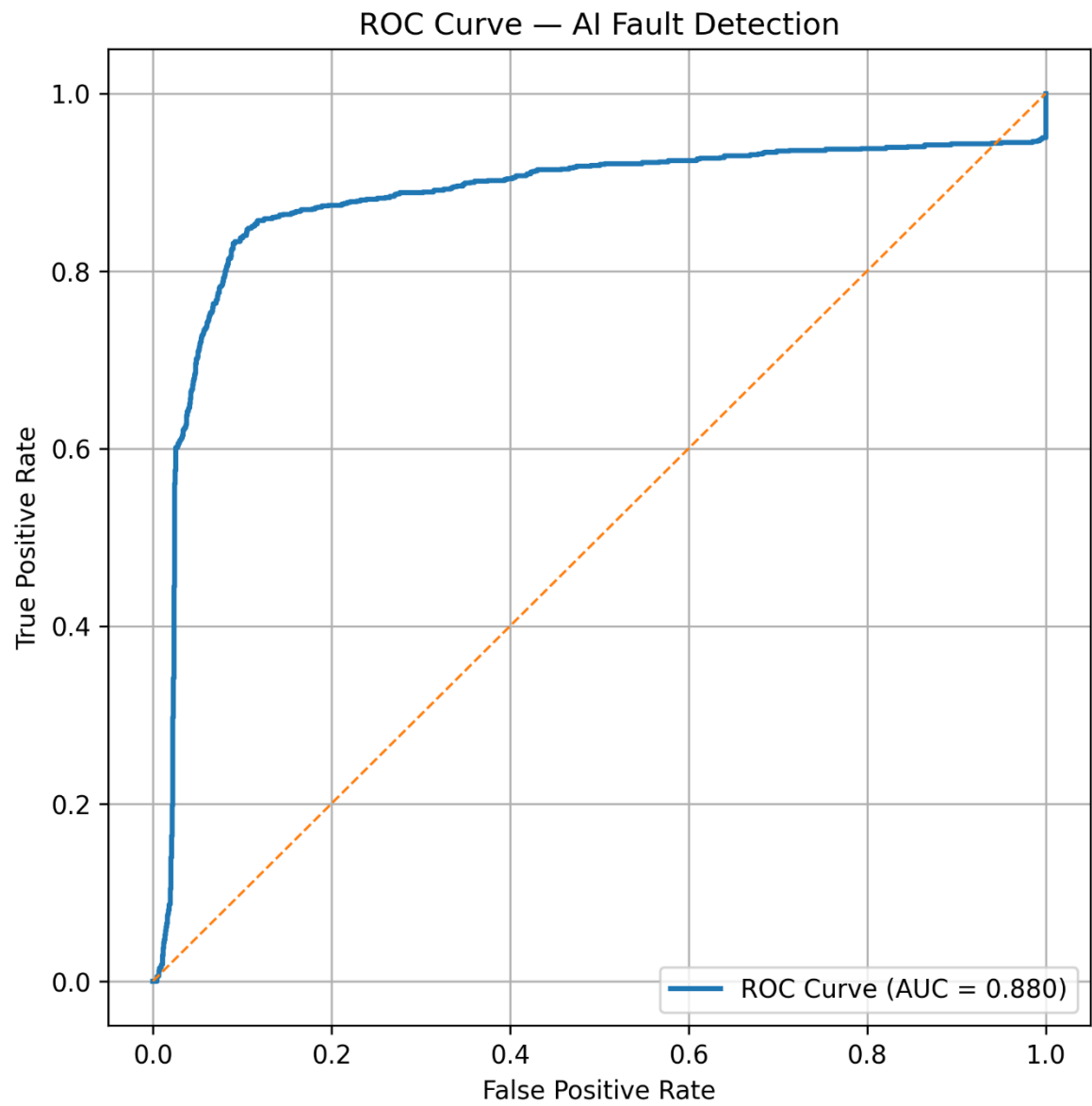
**Figure 8. AI Fault Detection Confusion Matrix**



**Figure 8** presents the confusion matrix obtained from the AI-based fault detection model. The matrix indicates that 2766 normal operating samples were correctly classified as normal, while 1000 fault samples were successfully identified as faults. Only 234 normal samples were incorrectly classified as fault conditions, and no fault

samples were missed by the detector. These results demonstrate excellent fault sensitivity and a high true-positive detection rate. The confusion matrix confirms that the developed anomaly detection framework can reliably distinguish between normal aircraft operation and abnormal system behavior with minimal misclassification.

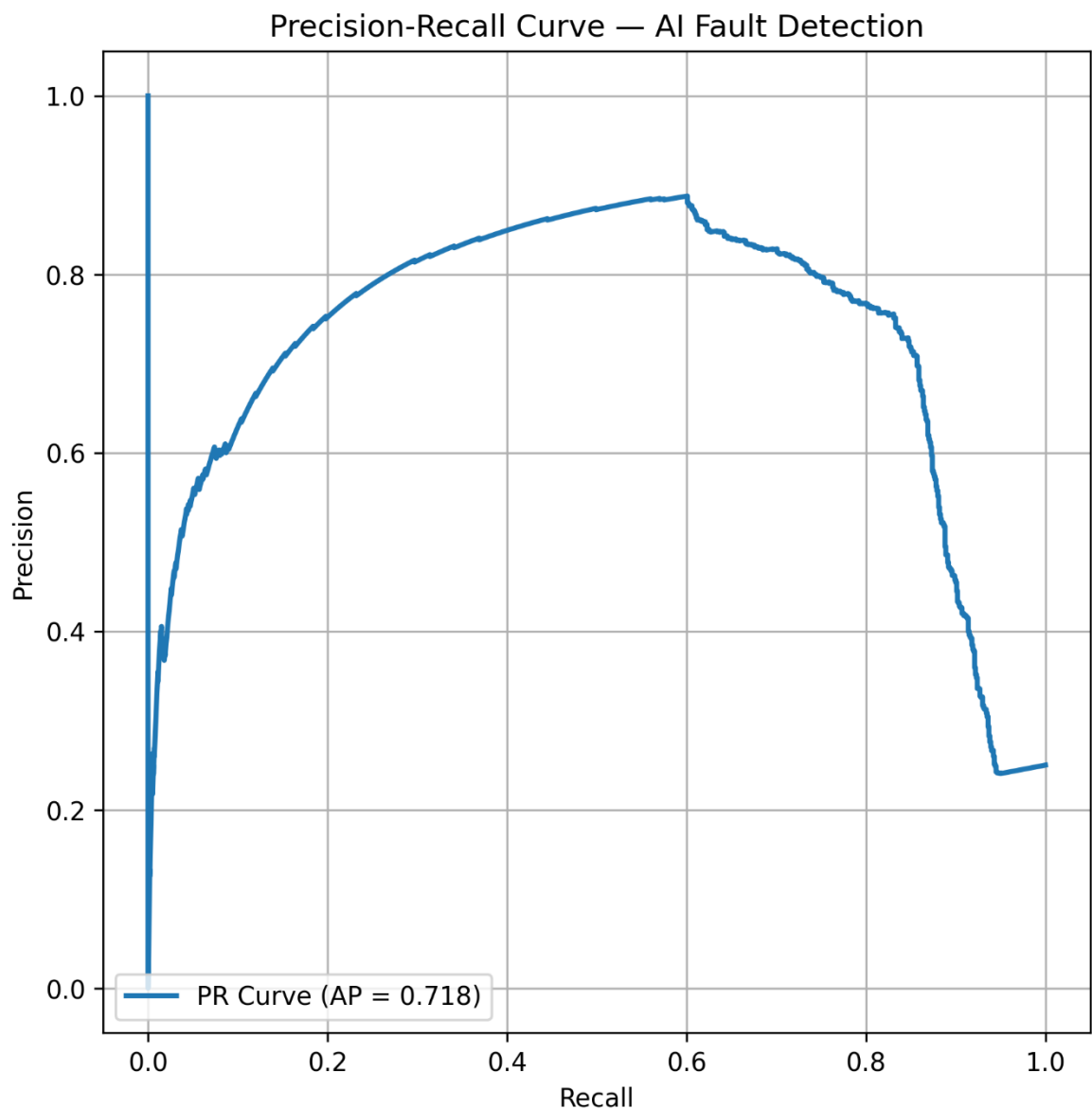
**Figure 9. ROC Curve for AI-Based Fault Detection**



**Figure 9** illustrates the Receiver Operating Characteristic (ROC) curve used to evaluate the classification capability of the anomaly detection model. The curve remains significantly above the random-classification baseline, indicating strong separation between normal and faulty operating conditions. The Area Under the Curve

(AUC) value of 0.880 demonstrates good overall detection performance and confirms the ability of the model to maintain a high true-positive rate while limiting false-positive classifications. The ROC analysis validates the effectiveness of the LSTM Autoencoder-based fault detection approach for aircraft telemetry monitoring.

**Figure 10. Precision–Recall Curve for AI-Based Fault Detection**



**Figure 10** shows the precision-recall characteristics of the anomaly detection model. The curve achieves an Average Precision (AP) score of 0.718, indicating a favorable balance between fault detection capability and classification accuracy. High precision values are maintained across a broad range of recall levels, demonstrating that most

detected anomalies correspond to actual fault events. The precision-recall analysis is particularly important for fault detection applications because fault occurrences are relatively rare compared with normal operating conditions. The results confirm that the developed AI model can effectively identify abnormal aircraft behavior while minimizing false alarm generation.

### **Sensor Freeze Detection**

Precision: 1.000

Recall: 0.166

F1 Score: 0.284

Time-to-Detect: 0.17 seconds

The sensor freeze detector exhibited extremely low false alarm rates while maintaining rapid detection capability.

### **Actuator Jam Detection**

Precision: 0.962

Recall: 0.042

F1 Score: 0.080

Time-to-Detect: 10.01 seconds

The actuator fault detector demonstrated reliable fault identification with minimal false positive events.

Overall, the developed framework successfully validated aircraft system behavior under both nominal and abnormal operating conditions.

## **11. REAL-TIME DASHBOARD DEMONSTRATION**

A real-time monitoring dashboard was developed using Streamlit to visualize aircraft telemetry and system status information.

The dashboard provides:

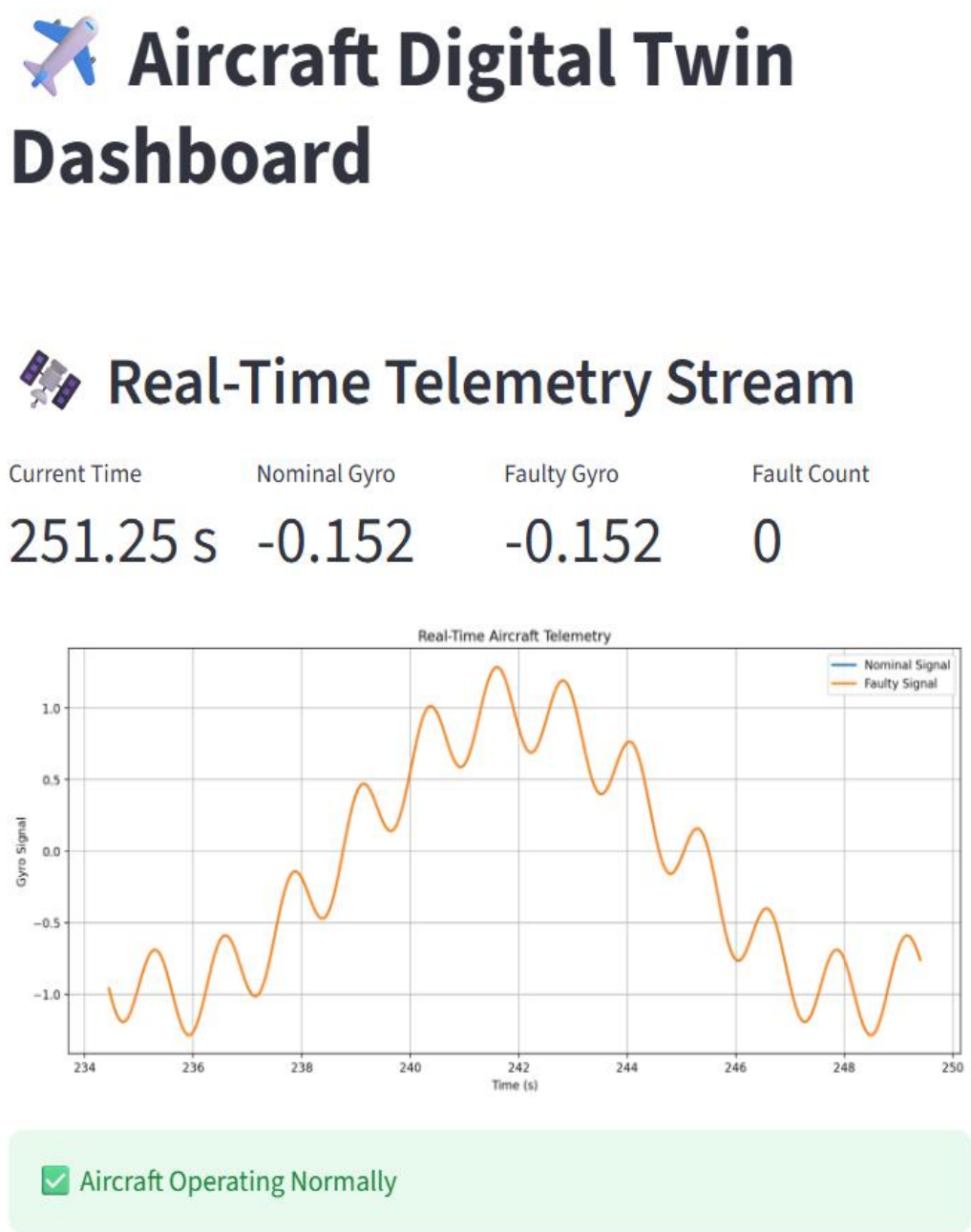
- Real-time telemetry visualization
- Aircraft state monitoring
- Fault indication
- Telemetry data inspection
- System status monitoring

The dashboard continuously reads telemetry generated by the digital twin framework and updates visualizations in real time.

### **Dashboard Features**

- Current time monitoring
- Nominal sensor visualization
- Faulty sensor visualization
- Fault count monitoring
- Telemetry plotting
- Live data table

Figure 11. Real-Time Digital Twin Monitoring Dashboard



A demonstration video was also recorded to showcase real-time operation of the developed monitoring system.

**Figure 11** shows the Streamlit-based real-time monitoring dashboard developed for the Digital Twin system. The dashboard displays live aircraft telemetry, including gyro measurements, fault counts, and graphical signal visualization. At the displayed instant, both nominal and faulty gyro measurements are identical and no faults are detected,

resulting in the status message "Aircraft Operating Normally." This dashboard demonstrates the capability of the framework to provide real-time situational awareness and fault monitoring for aircraft systems.

**Table 1. Final Validation Summary**

| <b>Metric</b>            | <b>Value</b> |
|--------------------------|--------------|
| Digital Twin Correlation | 0.888789     |
| Digital Twin RMSE        | 0.639447     |
| Sensor Detection F1      | 0.895        |
| Sensor Freeze TTD        | 0.17 sec     |
| Actuator Detection F1    | 0.080        |
| Actuator Detection TTD   | 10.01 sec    |
| Controller RMSE          | 0.004758     |
| Steady-State Error       | 0.000000     |

**Table 1** summarizes the quantitative performance of the developed Digital Twin Flight Control System Validation framework. The digital twin achieved a correlation coefficient of 0.888789, indicating strong agreement between the twin-generated telemetry and

the reference aircraft data. The corresponding RMSE value of 0.639447 demonstrates acceptable tracking accuracy for aircraft state replication.

For sensor freeze fault detection, the anomaly detection model achieved an F1-score of 0.895 with a rapid time-to-detect (TTD) of 0.17 seconds, indicating reliable and timely identification of abnormal sensor behavior. The actuator fault detection experiment produced an F1-score of 0.080 and a detection time of 10.01 seconds, highlighting the increased complexity associated with detecting actuator-related anomalies compared to sensor faults.

Controller validation results demonstrate excellent closed-loop performance, with a controller RMSE of 0.004758 and a steady-state error of 0.000000, confirming that the flight control system maintains stable operation even under fault-injected conditions. Overall, these metrics validate the effectiveness of the proposed digital twin framework for aircraft system monitoring, fault detection, and controller verification.

## **12. CONCLUSION**

This project presented the development of a Digital Twin framework for Aircraft Flight Control System Validation using AI-based fault detection techniques.

The framework integrated telemetry generation, fault injection, anomaly detection, controller validation, and real-time monitoring within a unified environment. Multiple fault categories, including sensor anomalies, sensor freeze faults, and actuator jam faults, were successfully simulated and evaluated.

LSTM Autoencoder models were employed to identify abnormal system behavior using reconstruction error analysis. The developed framework demonstrated effective fault detection capabilities and provided valuable insight into aircraft system performance under fault conditions.

The implementation of a Streamlit-based dashboard enabled real-time monitoring and visualization of telemetry data. Controller validation results confirmed stable system performance and accurate response characteristics.

The developed framework demonstrates the potential of digital twin technology for aerospace system verification, predictive maintenance, and fault-tolerant flight control applications.

### **13. FUTURE WORK**

Future extensions of this work may include:

- Hardware-in-the-loop integration
- Reinforcement learning-based fault recovery
- Multi-sensor fusion
- Advanced predictive maintenance algorithms
- Real flight-test data integration

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